

ENTHUSIAST ADVANCE COURSE

PHASE : MED-I (B)

TARGET : PRE-MEDICAL 2024

Test Type : MINOR

Test Pattern : NEET (UG)

TEST DATE : 18-06-2023

ANSWER KEY

Q.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
A.	1	1	4	3	3	2	3	1	4	4	3	2	2	2	1	3	1	1	2	4	2	4	1	4	1	1	2	2	1	2
Q.	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60
A.	4	3	3	2	3	1	3	3	4	4	2	3	2	3	4	4	2	2	4	3	3	2	1	2	3	1	3	2	1	2
Q.	61	62	63	64	65	66	67	68	69	70	71	72	73	74	75	76	77	78	79	80	81	82	83	84	85	86	87	88	89	90
A.	2	4	2	2	2	2	3	1	3	3	1	1	4	2	2	4	1	3	2	1	3	2	1	3	1	3	1	1	2	2
Q.	91	92	93	94	95	96	97	98	99	100	101	102	103	104	105	106	107	108	109	110	111	112	113	114	115	116	117	118	119	120
A.	2	2	4	2	2	4	2	2	4	4	1	2	2	2	3	4	4	3	2	1	3	4	1	1	2	1	3	4	3	2
Q.	121	122	123	124	125	126	127	128	129	130	131	132	133	134	135	136	137	138	139	140	141	142	143	144	145	146	147	148	149	150
A.	1	3	1	4	4	1	2	3	1	2	1	3	2	3	1	4	1	4	2	3	2	2	1	1	1	4	1	3	4	2
Q.	151	152	153	154	155	156	157	158	159	160	161	162	163	164	165	166	167	168	169	170	171	172	173	174	175	176	177	178	179	180
A.	3	3	2	3	1	2	2	2	3	4	4	2	2	3	2	3	2	4	3	3	3	3	3	3	4	2	4	1	3	3
Q.	181	182	183	184	185	186	187	188	189	190	191	192	193	194	195	196	197	198	199	200										
A.	2	2	1	2	1	2	4	3	1	3	4	2	4	4	4	4	1	2	2	3										

HINT - SHEET**SUBJECT : PHYSICS****SECTION-A**1. **Ans (1)**

By Gauss's theorem,

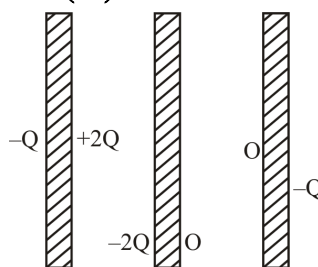
$$\phi = \frac{\Sigma q}{\epsilon_0}$$

Here $\Sigma q = 2e$

$$\Rightarrow \phi = \frac{2e}{\epsilon_0}$$

2. **Ans (1)**Electric field due to an arc = $\frac{2K\lambda}{r} \sin \frac{\alpha}{2}$ 3. **Ans (4)**

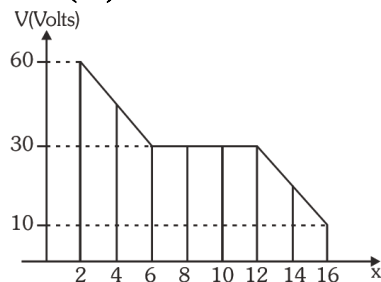
$$f \propto \frac{1}{r^4}$$

4. **Ans (3)**

After charge redistribution, final charges will be

 $\rightarrow$ So $x = +2Q$ & $x' = 0$ 5. **Ans (3)** $\phi_{g,s}$ is due to enclosed charge only but $E_{g,s}$ is due to all the charges.6. **Ans (2)**

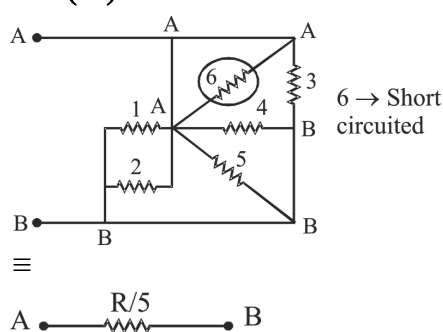
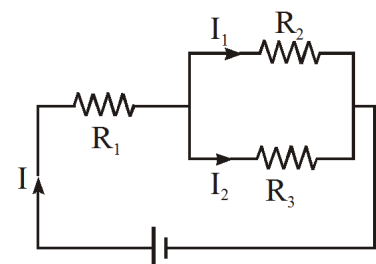
Electric field lines are always perpendicular to the equipotential surfaces. Also potential decreases in the direction of electric field.

7. **Ans (3)**

$$E = -\frac{dV}{dx} \text{ (-ve slope of V-x curve)}$$

electric field at 13m is

$$= \frac{30 - 10}{16 - 12} = \frac{20}{4} = 5 \text{ (V/m)}$$

11. **Ans (3)**12. **Ans (2)**

Since All resistance are equal

Power in R_1 will be $I^2 R_1$ which will be greatest.as $I > I_1 = I_2$

Hence option (2) is correct.

13. **Ans (2)**

From the given circuit,

$$V_A - (6 \times 2) - 12 - (9 \times 2) + 4 - (5 \times 2) = V_B$$

$$\text{Or } V_A - 12 - 12 - 18 + 4 - 10 = V_B$$

$$\text{Or } V_A - V_B = 48 \text{ volt}$$

14. **Ans (2)**

$$i = eA V_d, \quad i = \frac{dq}{dt} \quad 6t + 5 = 17$$

putting the values, $v_d = 1.77 \times 10^{-5} \text{ m/s}$.15. **Ans (1)**

Capacitor does not depends on charge & voltage

16. **Ans (3)**

$$\frac{V}{Q} = \frac{1}{C} = \text{slope}$$

$$\text{slope} \propto \frac{1}{C}$$

$$\text{so } C_B > C_A$$

17. **Ans (1)**

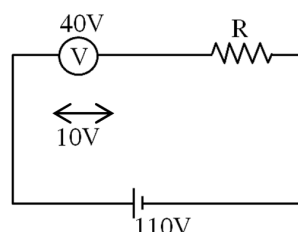
$$C = \frac{\epsilon_0 A}{d} \quad \& \quad C' = \frac{\epsilon_0 (2A)}{d/2} = \frac{4\epsilon_0 A}{d} = 4C$$

18. **Ans (1)**

$$C = K 4 \pi \epsilon_0 \frac{ab}{(b-a)}$$

$$= \frac{6}{9 \times 10^9} \frac{50 \times 60 \times 10^{-4}}{10 \times 10^{-2}}$$

$$= 2 \times 10^{-9} \text{ farad}$$

21. **Ans (2)**

$$I = \left(\frac{110}{40 + R} \right)$$

$$= \frac{110 \times 40}{(40 + R)}$$

$$40 + R = 440$$

$$R = 400 \text{ k}\Omega$$

23. **Ans (1)**

$$\text{Magnification } M = \frac{V_0}{u_0} \left(1 + \frac{D}{f_e} \right)$$

$$\text{for objective } \frac{1}{V_0} - \frac{1}{u_0} = \frac{1}{f_0}$$

$$\Rightarrow \frac{1}{V_0} - \frac{1}{(-5)} = \frac{1}{4}$$

$$\Rightarrow V_0 = 20 \text{ cm}$$

$$M = \frac{20}{5} \left(1 + \frac{20}{10} \right) \Rightarrow M = 12$$

25. **Ans (1)**

$$L = v_0 + f_e$$

$$v_0 = L - f_e = 75 - 5 = 70 \text{ cm}$$

$$\frac{1}{v_0} - \frac{1}{u_0} = \frac{1}{f_0}$$

$$\frac{1}{u_0} = \frac{1}{v_0} - \frac{1}{f_0} = \frac{1}{70} - \frac{1}{50} = -\frac{20}{3500}$$

$$u_0 = -175 \text{ cm}$$

29. Ans (1)

Here $f_a = -15\text{cm}$, $\mu_g = 1.5$, $\mu_l = 1.7$, $f_l = ?$

For the lens place in air:

$$\frac{1}{f_a} = (\mu_g - 1) \left[\frac{1}{R_1} - \frac{1}{R_2} \right]$$

$$\text{or } \frac{1}{-15} = (1.5 - 1) \left[\frac{1}{R} + \frac{1}{R} \right] = \frac{1}{R}$$

For the lens immersed in liquid :

$$\frac{1}{f_l} = \left(\frac{\mu_g}{\mu_l} - 1 \right) \left[\frac{1}{R_1} - \frac{1}{R_2} \right]$$

$$= \left(\frac{1.5}{1.7} - 1 \right) \left(\frac{1}{R} + \frac{1}{R} \right) = -\frac{0.2}{1.7} \times \frac{2}{R}$$

$$= -\frac{0.2}{1.7} \times \frac{2}{-15} \quad \text{or } f_l = +63.75 \text{ cm}$$

30. Ans (2)

Light rays passing from μ_1 also pass through μ_2 and they form one image similarly light rays passing through μ_2 also pass through μ_3 and they also form one image.

SECTION-B**36. Ans (1)**

$$E = \frac{\sigma}{\epsilon_0}$$

$$E_1 = E_2$$

$$\therefore \sigma_1 = \sigma_2$$

$$V = \frac{\sigma R}{\epsilon_0}$$

or $V \propto R$ (as $\sigma \rightarrow$ same)

$$\therefore \frac{V_1}{V_2} = \frac{R_1}{R_2} = \frac{a}{b}$$

37. Ans (3)

$$\frac{E}{r} = \tan 53^\circ \Rightarrow \frac{E}{3} = \frac{4}{3}, E = 4$$

$$\frac{kQ}{R^2} = 4, (R = 3)$$

$$\Rightarrow kQ = 36, Q = \frac{36}{k} = 144\pi\epsilon_0$$

38. Ans (3)

$$E = \frac{3k\lambda}{r} \sin\left(\frac{\pi/3}{2}\right) - \frac{2k(\sqrt{2}\lambda)}{2r} \sin\left(\frac{90}{2}\right)$$

$$= \frac{k\lambda}{r} - \frac{k\lambda}{r} = 0$$

40. Ans (4)

Potential difference across PQ i.e. p.d. across the resistance of 20Ω , which is $V = i \times 20$

$$\text{and } i = \frac{48}{(100 + 100 + 80 + 20)} = 0.16\text{A}$$

$$\therefore V = 0.16 \times 20 = 3.2\text{V}$$

41. Ans (2)

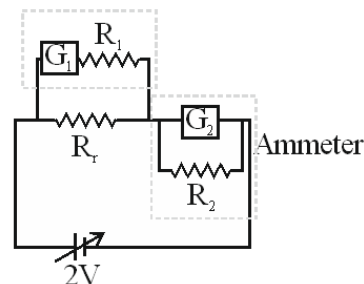
For ammeter $S = \left[\frac{I_g}{(I - I_g)} \right] G$. Therefore

$$\frac{G}{S} = \frac{I}{I_g} - 1. \text{ This gives } I = 2I_g \text{ for } G = S = 30\Omega.$$

New range will be $4I_g$, which will require $S = 10\Omega$. It can be obtained by shunting earlier shunt of 30Ω with a new shunt of 15Ω

42. Ans (3)

G_1 is used as a voltmeter by connecting it in series with high resistance R_1 and connected it across R_t in parallel and G_2 is used as an ammeter by connecting it in parallel with small resistance R_2 (called shunt) and connecting in series with R_t as shown in the figure.



Hence, above circuit is used to verify Ohm's law.

43. Ans (2)

$F = QE$, E shall be of one plate only.

46. Ans (4)

The far point of 6.0 m tell us that the focal length of the lens is $f = -6.0 \text{ m}$, $u = -18 \text{ m}$ and $h = 2 \text{ m}$.

$$\text{Using, } \frac{1}{f} = \frac{1}{v} - \frac{1}{u}$$

$$\Rightarrow \frac{1}{v} = \frac{1}{f} + \frac{1}{u} = \frac{1}{-6.0} - \frac{1}{18.0}$$

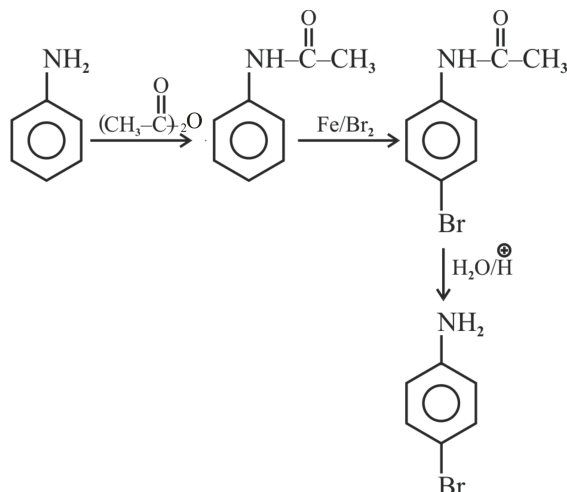
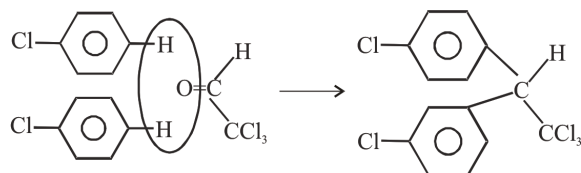
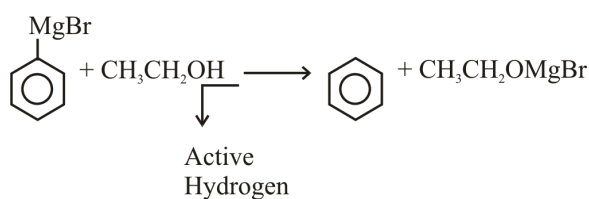
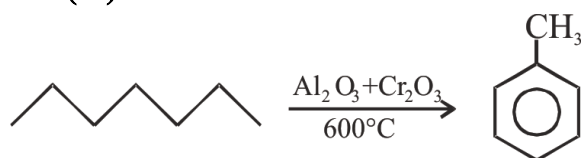
$$\Rightarrow v = -4.5 \text{ m}$$

$$\therefore \text{The image size, } h' = h \left(\frac{-v}{u} \right)$$

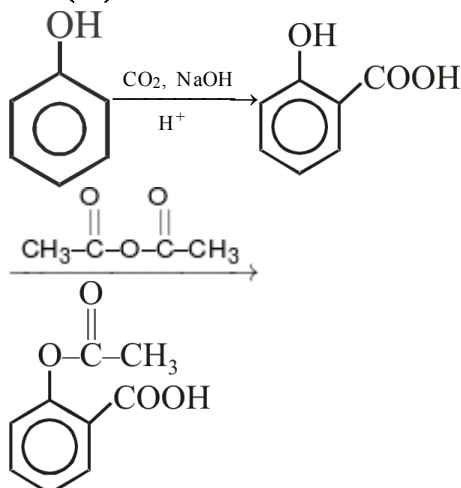
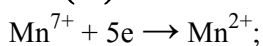
$$= 2 \times \left(\frac{-4.5}{18.0} \right) = 0.50 \text{ m}$$

SUBJECT : CHEMISTRY**SECTION-A**51. **Ans (3)**

Kolbe-Schmidt reaction.

52. **Ans (2)**Rate of $\text{ArS}_\text{N}2 \propto -\text{I} / -\text{M}$ group53. **Ans (1)**54. **Ans (2)**55. **Ans (3)**56. **Ans (1)**57. **Ans (3)**

Aniline does not undergo Friedel-Crafts reaction.

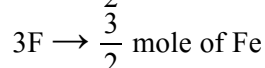
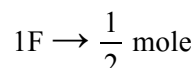
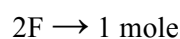
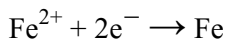
58. **Ans (2)**79. **Ans (2)**

Thus, 5 mole electron = 5 faraday.

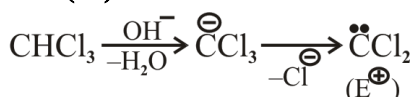
80. **Ans (1)**At equilibrium $\Delta G = 0 \Rightarrow E = 0$

$$0 = E^0 - \frac{0.059}{2} \log k_{\text{eq}}$$

$$\text{and } E^0 = 0.34 - (-0.25) = .59\text{V}$$

82. **Ans (2)**

$$= \frac{3}{2} \times 56 = 84 \text{ gm}$$

SECTION-B86. **Ans (3)**If E.W.G. is attached at O/P position of halo benzene then reactivity towards $\text{S}_\text{N}^\text{Ar}$ increases since it increases stability of CO.87. **Ans (1)**88. **Ans (1)**

NCERT Pg. # 358 Aldehyde ketones

89. **Ans (2)**

NCERT Pg. # 333 (para No. 2 & 3)

100. **Ans (4)**

$$E_{\text{cell}} = E_{\text{cell}}^{\circ} - \frac{0.059}{n} \log Q$$

$$\text{Slope of line} = - \frac{0.059}{n}$$

$$\text{Intercept} = E_{\text{cell}}^{\circ}$$

SUBJECT : BIOLOGY-I**SECTION-A**101. **Ans (1)**

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104. **Ans (2)**

NCERT Pg. # 108

105. **Ans (3)**

NCERT Pg. # 110

106. **Ans (4)**

NCERT Pg. # 106

107. **Ans (4)**

NCERT-XII, Pg. # 37

108. **Ans (3)**

NCERT-XII, Pg. # 36

109. **Ans (2)**

NCERT-XII, Pg. # 35

110. **Ans (1)**

NCERT-XII, Pg. # 37

111. **Ans (3)**

NCERT-XII, Pg. # 38

112. **Ans (4)**

NCERT Pg # 100

113. **Ans (1)**

NCERT-XII, Pg. # 102

114. **Ans (1)**

NECRT Pg # 101

115. **Ans (2)**

NCERT Pg # 101

116. **Ans (1)**

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117. **Ans (3)**

NCERT- Pg. # 101, 102

118. **Ans (4)**

NCERT-XII, Pg. # 101-102

119. **Ans (3)**

NCERT-XI, Pg. # 303

120. **Ans (2)**

NCERT-XII, Pg. # 102

121. **Ans (1)**

NCERT-XI, Pg. # 101

122. **Ans (3)**

NCERT-XI, Pg. # 102

123. **Ans (1)**

NCERT XIIth, Pg.No.103 Q-1 7.1.2

124. **Ans (4)**

NCERT Pg. # 102

125. **Ans (4)**

NCERT Pg # 101

126. **Ans (1)**

NCERT Pg. # 102, 103

127. **Ans (2)**

NCERT Pg # 103

128. **Ans (3)**

NCERT-XII, Pg. # 102

129. **Ans (1)**

NCERT-XI, Pg. # 101

130. **Ans (2)**

NCERT Pg # 101

131. **Ans (1)**

NCERT PAGE (ENGLISH): 101

132. **Ans (3)**

NCERT Pg # 100

133. **Ans (2)**

NCERT Pg # 102-103

134. **Ans (3)**

NCERT PAGE (ENGLISH): 100,101

135. **Ans (1)**
NCERT Pg#147

SECTION-B

136. **Ans (4)**
NCERT Pg. # 110

137. **Ans (1)**
NCERT Pg. # 109, 107

138. **Ans (4)**
NCERT Pg. # 106

139. **Ans (2)**
NCERT-XII, Pg. # 36

141. **Ans (2)**
NCERT Pg. # 104

142. **Ans (2)**
NCERT-XII, Pg. # 103

143. **Ans (1)**
NCERT Pg. # 106

144. **Ans (1)**
NCERT Pg # 104-105

145. **Ans (1)**
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146. **Ans (4)**
NCERT- Pg. # 103

147. **Ans (1)**
NCERT-XII, Pg.# 104

148. **Ans (3)**
NCERT Pg. # 103

149. **Ans (4)**
NCERT Pg. # 104

150. **Ans (2)**
NCERT PAGE (ENGLISH): 100

SUBJECT : BIOLOGY-II

SECTION-A

151. **Ans (3)**
NCERT PAGE (ENGLISH): 147, 152, 154, 157

152. **Ans (3)**
NCERT-XI, Pg#150,151(E), 162,163(H)

153. **Ans (2)**
NCERT-XII Pg. # 151, 152

154. **Ans (3)**
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155. **Ans (1)**
NCERT-XII, Pg # 92

157. **Ans (2)**
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158. **Ans (2)**
NCERT-XII, Pg. # 158, 159

159. **Ans (3)**
NCERT-XII, Pg. # 164, 153

174. **Ans (3)**
NCERT (XI) Pg. # 144

175. **Ans (4)**
NCERT (XIth) Pg. # 152

178. **Ans (1)**
NCERT (XII) Pg. # 97

SECTION-B

186. **Ans (2)**
NCERT Pg # 101

187. **Ans (4)**
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188. **Ans (3)**
NCERT Pg # 103

189. **Ans (1)**
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190. **Ans (3)**
NCERT - XII

191. **Ans (4)**
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192. **Ans (2)**
NCERT XII Page No. # 159/158 (E), 173/174 (H)

197. **Ans (1)**
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